

End of Topic Test – Chapter 5: The Normal Distribution (Test 2)

Answer all questions in the spaces provided. A standard normal distribution table $\Phi(z)$ is provided in the Appendix.

1. Understanding the normal distribution

A bakery finds that the masses of the loaves of bread it bakes are modelled as being normally distributed with mean 800 g and standard deviation 15 g.

- (a) Sketch a labelled normal distribution curve for the mass of a loaf, showing the mean and one standard deviation on each side. (2 points)
- (b) State what proportion of loaves are expected to have a mass between 785 g and 815 g. (2 points)
- (c) Explain why the normal distribution is a suitable model for the mass of a loaf of bread. (2 points)

2. Calculating probabilities

The volume of coffee, X ml, dispensed by a machine into a cup is modelled by $X \sim N(340, 16^2)$.

(a) Calculate $P(X > 365)$. (3 points)

(b) Calculate $P(310 < X < 350)$. (3 points)

(c) Interpret your answer to part (b) in context. (2 points)

3. A job screening test

An aptitude test used by an employer produces scores that are modelled by $X \sim N(58, 11^2)$.

(a) Find the probability that a randomly selected candidate scores under 45. (3 points)

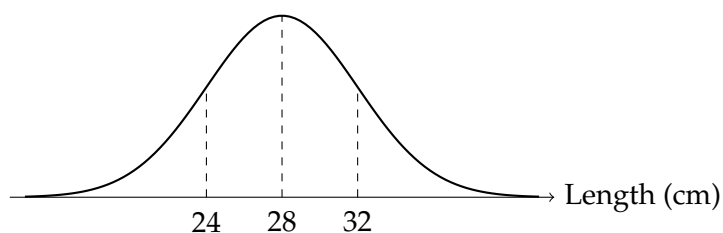
(b) Candidates scoring in the top 20% of scores are invited to interview. You are given that, for the standard normal distribution, $P(Z > 0.8416) = 0.20$. Find the minimum score needed for an interview invitation. (3 points)

(c) Explain why the normal distribution may not perfectly model aptitude test scores in practice. (2 points)

4. Interpreting a normal distribution diagram

A biologist measures the lengths of fish caught in a lake. The lengths are approximately normally distributed with mean 28 cm and standard deviation 4 cm.

The diagram below shows a simplified normal curve.



- (a) What proportion of fish are expected to have a length between 24 cm and 32 cm? (2 points)
- (b) Estimate the proportion of fish with a length greater than 38 cm. (3 points)
- (c) Give one reason why the biologist might still use the normal model even though real fish-length data may be slightly skewed. (2 points)

5. Interpreting a study using the normal distribution

A cognitive psychology study measured simple reaction times (in milliseconds) for a large sample of adults performing a visual detection task. The researchers reported that reaction times were “approximately normally distributed with a mean of 250 ms and a standard deviation of 30 ms.” The study also noted:

“While the normal model captures the overall pattern well, individual reaction times are affected by factors such as momentary lapses in attention, fatigue, and practice effects, which can produce occasional unusually slow or unusually fast responses that are not well predicted by the model.”

The authors added that these occasional extreme responses – both very fast anticipatory responses and very slow lapses – occurred more often than a strict normal model would predict, and recommended caution when using the model to estimate the probability of very extreme reaction times.

- (a) State the meaning of “normally distributed” in this context. (2 points)

- (b) Using the model with mean 250 ms and standard deviation 30 ms, calculate the proportion of adults expected to have a reaction time above 305 ms. (3 points)

- (c) Explain how the quotation challenges the assumption that the normal model perfectly represents real reaction-time data. (3 points)

- (d) Suggest one reason why extreme reaction times may be less reliable than reaction times near the mean. (2 points)

END OF TEST

Appendix: Standard Normal Distribution Table

The table gives values of $\Phi(z) = P(Z < z)$ for the standard normal distribution $Z \sim N(0, 1)$, for $z \geq 0$. For negative values of z , use $\Phi(-z) = 1 - \Phi(z)$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990